

Rezaur Rahman Ratul

AI Researcher | Medical Imaging · LLMs & Conversational AI · ML Engineer

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EDUCATION

United International University (UIU)

B.Sc. in Computer Science and Engineering

Relevant Areas: Machine Learning, Deep Learning, Medical Imaging, Computer Vision, Healthcare AI, LLMs.

Dhaka, Bangladesh

Jul 2021 – Expected Jun 2026

SUMMARY

I am a final-semester CSE student at UIU with research experience in healthcare AI, medical imaging, deep learning, and large language models. I am a published author at NSysS conference 2025 (XAI & Conversational AI), thesis **Champion** for voice reconstruction in stroke patients, and a builder of award-winning clinical prototypes motivated to contribute AI-driven solutions for early disease detection and rural healthcare access.

RESEARCH EXPERIENCE

Voice Reconstruction from Lip Movements for Stroke Patients

2025 – Present

- Designed a deep learning pipeline for visual speech reconstruction using computer vision and signal processing; evaluated against healthcare-grade benchmarks.
- Champion**, UIU Final Year Design Project competition; submitted to a peer-reviewed venue.

Image Segmentation and Classification – Waste Dataset

2025 – Present

- Applied ensemble segmentation models to real-world multi-class image data; conducted ablation studies and model evaluation – directly transferable to medical imaging pipelines.

Web Security Vulnerability Detection

Jun 2025 – Present

- Built a large-scale SQL injection dataset (>1M samples) and designed automated detection frameworks covering diverse attack categories.

EXPERIENCE

University Innovation Hub Program (UIHP) - Trainee

Dhaka, Bangladesh Nov 2024 – Mar 2025

- Founding member of **SHOHOCHOR** (Developed a prototype CCTV-based child monitoring solution for daycare centers, leading system planning and team validation to enhance child safety and real-time supervision. **Top 4** nationally, won 20,000 BDT.

PUBLICATION

An Explainable and Scalable Framework for Compost Maturity Prediction Using Ensemble Learning and Conversational AI.

M. A. O. Mullick, A. H. R. A. Mufti, Rezaur Rahman Ratul, J. Noor. *NSysS 2025*, pp. 29–34. [ACM Digital Library](#)

(XAI · Conversational AI · Ensemble DL · Scientific Writing)

PROJECTS

Astha: Elderly Health Monitoring Application

- End-to-end healthcare app: blood pressure, heart rate, health records, caregiver alerts, and early risk detection. **1st Runner-up**, UIU System Analysis Design Lab Competition.

IntelliSense: AI-Assisted Smart Wheelchair for Paralysis Patients

Award

- Gesture/voice-controlled wheelchair with pulse monitoring, fall detection, obstacle avoidance, and medication reminders. **1st Runner-up**, UIU Electronics Lab Competition.

Landlytics: Intelligent Land Analytics Platform

GitHub

- Laravel platform for agricultural analysis, crop recommendation, weather monitoring, and data-driven land planning.

SKILLS

Machine Learning Supervised & Deep Learning, Model Training & Fine-tuning, Medical Image Processing, Image Segmentation, Feature Extraction

AI Domains Healthcare AI, Explainable AI (XAI), Large Language Models (LLMs), Conversational AI, Speech & Signal Processing, Assistive Technology

Programming Python, PHP, JavaScript, SQL, HTML, CSS

Frameworks PyTorch, TensorFlow, Git, Jupyter Notebook, Laravel

Research Experimental Design, Data Preprocessing, Prototype Development, Scientific Writing, Literature Review

AWARDS, ACHIEVEMENTS & CERTIFICATIONS

Champion : Final Year Design Project, Voice Reconstruction (UIU) **1st Runner-up** : System Analysis Design Lab, Astha (UIU)

1st Runner-up : Electronics Lab, IntelliSense Wheelchair (UIU) **Top 4** : UIHP SHOHOCHOR Healthcare Innovation (National)

Top 10 : Harvard Hackathon Bangladesh

Top 30 : National Blockchain Competition

Certifications: Introduction to Generative AI · Basics of AI · Cybersecurity Essentials · CCNA: Introduction to Networks · Introduction to Cybersecurity

REFERENCES

Dr. Mohammad Nurul Huda

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